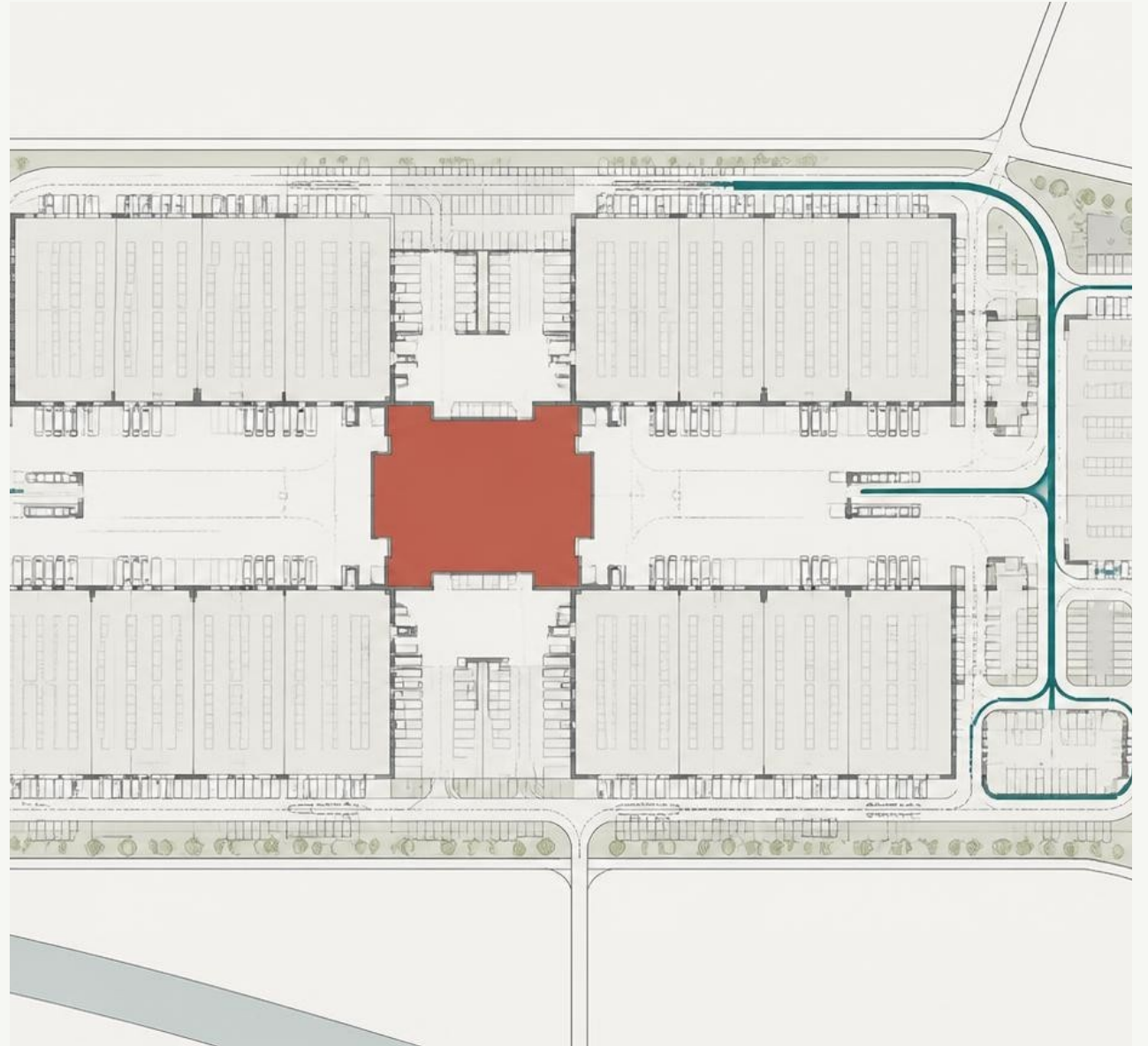


E-Commerce Sales & Customer Analytics

Turning transaction data into
business insights with SQL

Tools: SQL, DuckDB, VS Code, Git, GitHub
Dataset: 1,200 orders, January 2023 to June
2025



From raw transactions to business intelligence

01 Performance Monitor sales run rates, monthly volume, and aggregate revenue trends.	02 Customer behavior Identify repeat buyers, segment by value, and analyze retention gaps.	03 Product analytics Evaluate unit volumes, catalog distribution, and revenue contributions.
04 Marketing impact Measure channel effectiveness, referral sources, and coupon utilization.	05 Data-driven decisions Translate SQL queries into actionable targets and operational priorities.	

A complete analytical workflow moving from raw data extraction to strategic business intelligence.

Focus areas: sales, retention, products, marketing, and revenue trends.

1,200 orders across 2.5 years

Measure	Value
Orders	1,200
Customers	1,189
Total revenue	\$1.26M
Average order value	\$1,053.97
Period	2.5 years (Jan 2023 – Jun 2025)

Data covers order details, customers, products, quantity, unit price, revenue, payment method, coupon code, referral source, and order status.

Source: Transaction database snapshot, 2023-2025



Five business questions drive the analysis

01 **Sales:** How much revenue is generated and how does it change over time?

02 **Customers:** Who are the highest-value customers and how strong is repeat purchasing?

03 **Products:** Which products drive sales and revenue?

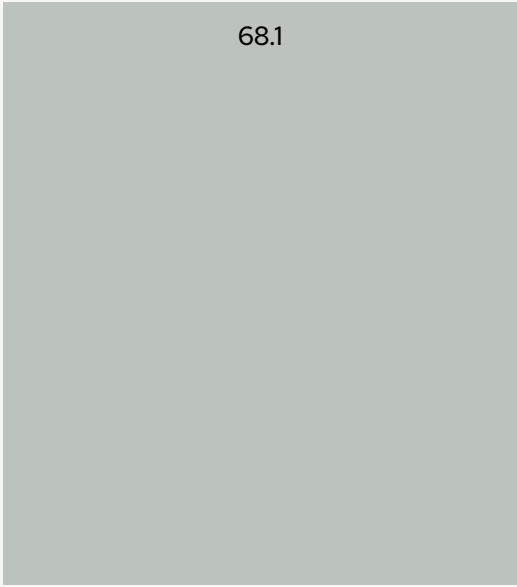
04 **Marketing:** Which acquisition channels and coupons perform best?

05 **Operations:** How are payment methods and order statuses distributed?

The business at a glance: \$1.26M from 1,200 orders



May 2023



Jun 2024

\$1,053.97

Average order value

Total revenue reached \$1,264,761.96 across 1,189 unique customers. Revenue stayed relatively consistent over 2.5 years, peaking at \$68.1K in June 2024 and \$63.8K in May 2023.

Insight: High AOV indicates strong medium- to high-value purchases; customer retention is the prime growth opportunity.

Only 11 customers ordered more than once

11

Repeat customers out of 1,189

C38840

Top customer

\$5,723.23 total spend across 2 orders, with an AOV of \$2,861.62.

Most repeat customers placed exactly two orders. While single-order volume carries the business, high individual spend proves strong willingness to pay.

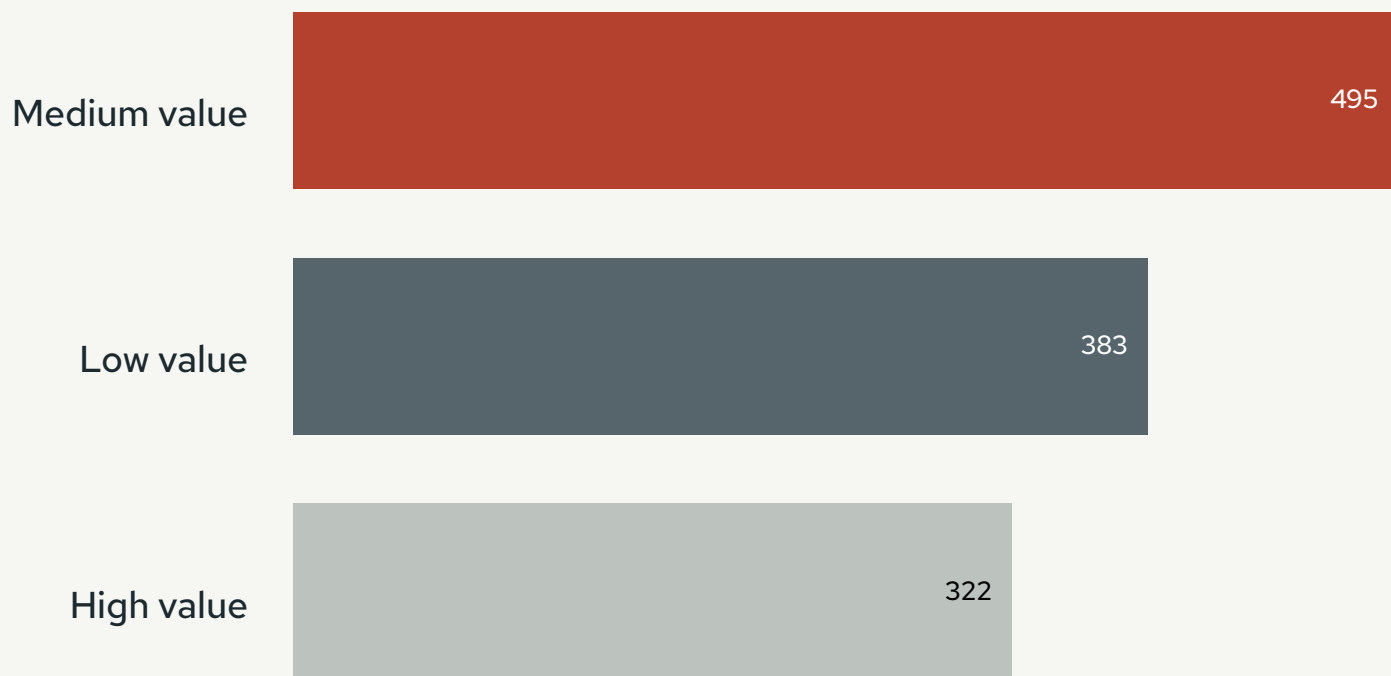
Source: Customer frequency SQL query, 2023-2025

34

High spenders

Customers who spent over \$3,000 in total across the catalog.

Customer segments by order count



Growth opportunities

Medium value forms the largest cohort with 495 orders, followed by low value at 383 and high value at 322.

- Customer retention
- Loyalty programs
- Personalized marketing
- Customer lifetime value

Segmented by total order value thresholds

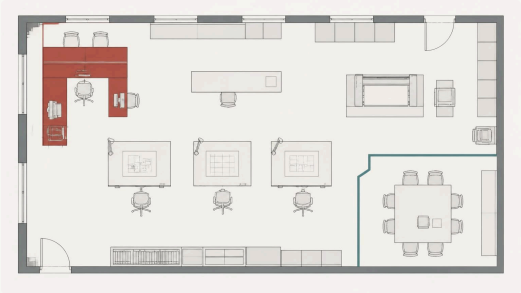
Chair, Printer and Laptop lead on volume and revenue



Unit sales leaders

Chair led volume with 562 units, followed closely by Printer at 542, Laptop at 535, and Phone at 411 units.

Insight: The top three products earn nearly identical revenue, each generating over \$192K.



Revenue is spread evenly across seven products

15.47%

Chair & Printer share

Each contributed ~15.47% of total sales, representing top individual revenue shares.

15.19%

Laptop share

Laptops accounted for 15.19% of sales, completing the core trio.

\$1,110

Laptop AOV

Highest average order value across the catalog at \$1,110.56 per order.

\$1,000

+

Category baseline

Chair, Printer, and Monitor each maintained AOVs above \$1,000; Phone stood at \$972.58.

A balanced portfolio protects the business: there is zero dependency on any single flagship product.

Source: Product mix & pricing distribution SQL model, 2025

Instagram is the top channel; FREESHIP the top coupon

Acquisition channels

- Instagram: highest revenue \$275,285.45 and most orders, 259
- Email: \$261,808.55; Google and Facebook comparable; Referral lowest

Promotions & payments

- FREESHIP: \$335,036.99 from 313 orders; orders without coupons made \$322,401.41
- Credit Card: highest revenue \$263,847.63; Online Payments: most orders, 258